What Crosses the Boundary? Measuring the Retarget-to-Track Interface for Humanoid Tracking

Anonymous ICRA Submission

Abstract—**Humanoid retargeting pipelines usually hand trackers explicit robot-space targets, so a learned representation can be ignored. We make this interface measurable: a 64-d code is the decoder's exclusive motion command, while matched upstream controls vary what can cross it. A single simulated Unitree G1 walk exposes the confound: a frozen retargeting latent reaches 0.656 ± 0.044, below absolute time at 0.754 ± 0.010, so one deterministic clip cannot separate content from progress. We therefore preregister a two-walk assay with two passing specialists, 69 reference-only pairs matching present states but diverging in the future, and time, proprioception, and phase-matched other-trajectory controls. In the frozen three-seed aggregate, ambiguity completion is 0.754 for the retargeting latent, 0.562 for time, and 0.552 for shuffled content; paired 95% intervals for A–T and A–S are [0.124, 0.274] and [0.127, 0.279]. Destroying the seed-0 explicit student's exclusive command collapses completion from 0.925 to zero. The assay identifies control-usable trajectory information beyond absolute within-clip time, scoped to two walks, one simulated robot, one frozen representation, and one controller class.**

I. Introduction

Many modern humanoid motion systems use two stages: a retargeter converts human motion into robot-space joint trajectories, and a reinforcement-learning policy learns to track them [7]. The split is useful—retargeters and trackers can be developed, evaluated, and reused independently—and recent systems are strong on both sides: interaction-preserving optimization on the retargeting side [2], physics-in-the-loop retargeting [1], and unified tracking controllers that follow thousands of clips with one network [4], [5]. Yet the *interface* between these stages is rarely the object of study. What crosses the boundary is usually a per-frame stack of joint targets—a format rather than an explicitly modeled representation.

That format has three limitations. First, it does not explicitly expose information beyond the joint trajectory, such as contact reasoning, uncertainty, or structure shared across embodiments. Second, kinematic metrics such as pose error, foot skate, and penetration do not directly measure *trackability*; the GMR study [3] shows that retargeting quality can bound downstream RL tracking. Third, the format is tied to an embodiment: a 29-DoF command stream cannot directly command a 23-DoF robot.

Simply concatenating the retargeter's latent beside the explicit reference fails in our baseline: the policy takes the more informative channel. Our latent-command RL baseline also collapses; recent systems instead use staged distillation or representation pretraining [6], [19]. What has been missing is not another repair but a measurement.

Our key insight is that exclusivity makes the interface experimentally measurable. If a learned code is the policy's only route to the goal, then changing the signal that feeds the code and measuring closed-loop completion tests what the controller can use. A deterministic single-trajectory test remains confounded because absolute time identifies the target. We build the instrument from SNMR, a retargeting network distilled from a classical IK teacher whose shared latent serves five humanoids, and a 64-d command code trained by online distillation from an explicit-command RL teacher [8]. The goal is visible to the code encoder and never to the action decoder (§III).

On a simulated Unitree G1 the learned interface matches the evaluated explicit-command teacher checkpoint on one walk (three student seeds, one teacher checkpoint), but the frozen retargeting latent loses to an absolute-time control. We turn that negative diagnostic into the two-walk paired assay in Fig. 1c. A world-body indexing defect found during calibration is kept as a validity audit: repairing it cut joint RMSE by 46%, and all interface results reported here were rerun after repair (§VII).

Contributions.

- **A causal interface instrument.** A 64-d command is the decoder's only goal channel; in the seed-0 explicit student, zero, shuffle, and matched-marginal interventions all collapse completion from 0.925 to 0.000, separating structural exclusivity from causal use.
- **A paired assay for content beyond time.** A policy-independent search yields 69 two-walk starts with similar present states and different futures. Under one frozen student recipe, SNMR passes the registered content gate over absolute time and phase-matched shuffled content.
- **Controls and reproducible failure boundaries.** The single-clip time null retires a favorable latent claim; passing specialist and explicit-student gates isolate representation from task feasibility; an audited five-humanoid retargeter, hashes, protocols, and negative-results ledger make those decisions reproducible.

These controls also falsified three earlier working hypotheses; we retain those outcomes in the negative-results ledger rather than selecting only favorable experiments.

II. Related Work

Retargeting as preprocessing. Classical retargeting solves per-frame IK with hand-tuned correspondences [3]; interaction-mesh methods add hard contact constraints [2]; ReActor moves retargeting inside physics via bilevel RL,

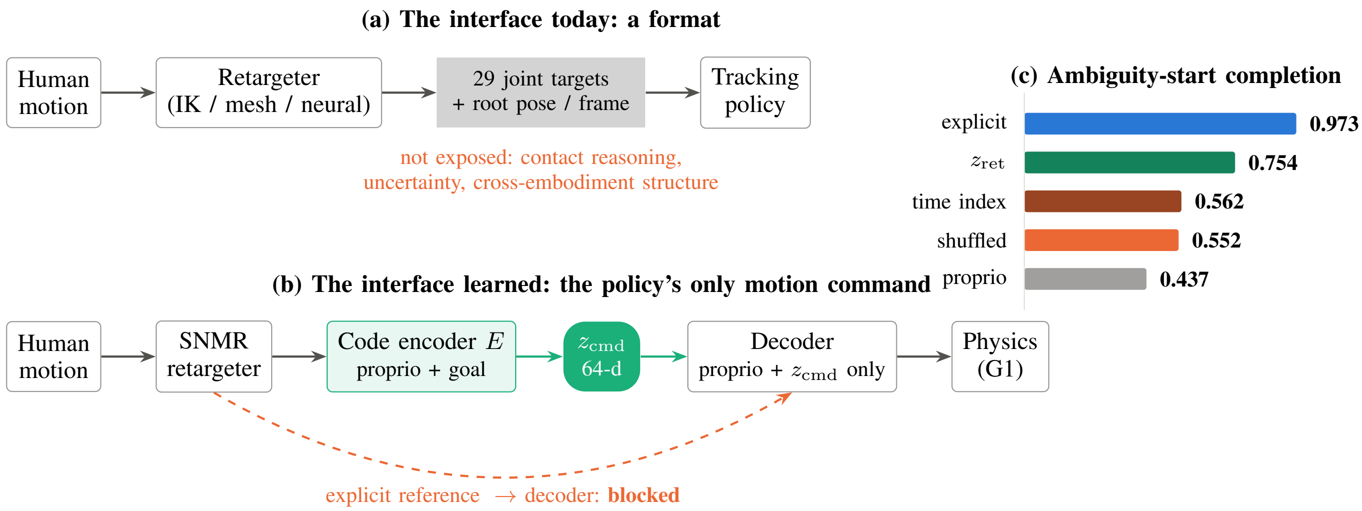

Fig. 1. **Measuring the retarget-to-track interface.** (a) The standard humanoid stack passes per-frame joint targets between retargeting and tracking without explicitly exposing the retargeter's contact reasoning, uncertainty, or cross-embodiment structure. (b) The instrument: a learned 64-d command code—the goal reaches the code encoder but *never* the decoder, so z_{cmd} is the policy's only motion command. (c) In the preregistered two-walk three-seed aggregate assay, the frozen retargeting latent passes the registered content gate over matched time and phase-shuffled controls on paired ambiguous starts (69 pairs; A–T $+0.191$, 95% CI $[0.124, 0.274]$).

eliminating artifacts by construction [1]. All three families emit the same interface—robot joint trajectories—and are evaluated by kinematic quality plus, recently, downstream RL success [3], [2]. These works do not treat the *representation* handed to the tracker as the experimental object. Neural feed-forward retargeters [16], [17], [18] amortize the optimization and, in AdaMorph's case, span many embodiments through a morphology-agnostic latent—but their latents are internal machinery rather than evaluated command interfaces.

Tracking and its command spaces. Modern trackers follow references specified as joint targets with body-pose terms [4], [5], [15], now at foundation scale [23], [24]; HOVER unifies multiple command *modes* by distillation [14]; UniTracker compresses the reference into a CVAE latent and is the closest precedent for our command interface—single-embodiment, with no retargeter and no representation analysis [6]. Latent skill spaces for physics characters [11], [12], [13] establish code-mediated control; our calibration ablation instead selects a deterministic encoder for the frozen assay. Behavioral foundation models [20] reach a promptable latent from the opposite entry point—unsupervised RL without a retargeting stage. Capability benchmarking in the HOVER/ULTRA class [14], [22] asks how broadly or accurately a controller can be commanded; that axis is orthogonal to representation measurement—what information crosses an exclusive interface and is used. RoboGhost bypasses explicit retargeting by conditioning control on a language-grounded motion latent [21], while ULTRA distills a tracker into a multimodal latent controller [22], AnyBody distills a tracker into a keypoint-addressable latent [25], and HANDOFF engineers a compact task-space command, arguing that the interface choice is decisive [26]. For dense tracking, however, ULTRA sends the full-body goal to the decoder through a residual shortcut, and all of these engineer the channel rather than measure it. Our exclusivity contract instead makes the learned code the only goal channel and tests a frozen multi-embodiment retargeting latent as one possible source for that code.

Interface critiques and measurement. BeyondMimic names a "planning-control gap" [4]; the GMR study quantifies how retargeting quality caps tracking [3]. Recent work sharpens the concern: a controlled study of explicit command signals finds proprioceptive history does not recover reference phase and velocity [27]; OmniTrack and GenTrack intervene on the boundary by replacing references with dynamics-consistent ones [28], [29]; and benchmark audits report that tracking metrics conflate motion difficulty with policy quality [30] and that hierarchical results are tracker-conditioned [31]. None makes the interface itself the object of a controlled causal measurement. Our instrument imports the missing methodology: probing classifiers and their documented gap between decodability and use [32], [33], interventions that ablate information and measure behavior rather than decode it [34], usable information under computational constraints [35], and preregistered endpoints [36], [37]; the single-trajectory clock of §IV is an instance of shortcut learning [38].

III. The Instrument: A Learned Code Under an Exclusivity Contract

A source motion m_t enters a retargeter R; the standard interface hands the tracker π a 64-d reference $g_t = (q_t^{\mathrm{ref}}, \dot{q}_t^{\mathrm{ref}}, r_t^{\mathrm{ref}})$: 29 joint positions, 29 velocities, and a 6-d relative reference-orientation representation. We move the actor-visible reference behind a learned code: $z_{\mathrm{cmd}} = E_\theta(x_t, u_t)$, where $u_t = g_t$ in the explicit arm and $u_t = [z_{\mathrm{ret},t}, z_{\mathrm{ret},t+0.1\,\mathrm{s}}]$ in the SNMR arm. The explicit command is current-frame only; an explicit future-goal window remains future work. Under an *exclusivity contract*, the upstream signal is visible to the encoder E_θ, but never to the action decoder, whose input is exactly the 90-d proprioceptive slice plus the 64-d code (structurally regression-tested). Causal use is established on the clean seed-0 two-walk explicit student: zeroing, batch-shuffling, or independently resampling each observed marginal of z_{cmd} collapses completion from 0.925 to 0.000. We use "interface," not "bottleneck": the code is not dimensionally narrower than the goal (64-d each), and no rate term constrains the goal-to-code path. Linear diagnostics find effective rank 14 of 64 and lower goal decodability from the deployed code than from proprioception (held-out R^2 0.48 vs 0.62); these characterize the representation but do not establish an information bound. The encoder's upstream source is the experimental factor (§IV). Across our diagnostics it can be the robot-space reference g_t, the source-side representation z_{ret} alone (with no robot-space reference anywhere in the student), both signals, absolute time, or proprioception alone. E70 freezes the five-arm comparison C/A/T/B/S defined in §IV.

Operational estimand. Let x_t be actor-visible proprioception, u_t the candidate upstream signal, E the learned command encoder, and D the action decoder:

$$z_t = E(x_t, u_t), \qquad a_t = D(x_t, z_t), \qquad u_t \not\to D \text{ otherwise.} \tag{1}$$

For rollout completion Y, the quantity of interest is not mutual information in z_{ret} but a contrast between policies produced by the fixed training algorithm:

$$\Delta_{A,T} = \mathbb{E}[Y(D_A(x, E_A(x, z_{\mathrm{ret}}))) - Y(D_T(x, E_T(x, \tau)))], \tag{2}$$

where τ is matched within-clip time. The arms share the teacher ensemble, core architecture, training budget, initialization seed, and evaluation starts; their closed-loop training states may differ. C verifies that the student class can transmit a sufficient goal; B measures what closed-loop proprioceptive memory can do without a goal; S supplies the other trajectory's latent at matched normalized time. Finally, interventions on the *learned* z_t test whether the structurally exclusive route is actually used. These controls replace the broad statement "the latent contains information" with the narrower operational claim we can test: under a frozen training and evaluation protocol, a policy trained from the latent achieves different physical task success from one trained from time. Only the within-policy z_{cmd} destruction tests are causal interventions on a fixed trained controller.

The latent's source is SNMR: a graph-attention encoder over the source skeleton (per-node features, adjacency, and global pooling) maps motion to a per-frame retargeting latent $z_{\mathrm{ret}} \in \mathbb{R}^{128}$ (distinct from the control code z_{cmd}); embodiment-conditioned graph decoders (AdaLN on a 32-d embodiment code, $\tanh$ joint-limit heads) emit trajectories for the five training humanoids. Trained by distillation from a per-frame IK teacher [3] on 77 LAFAN1 clips × 5 robots (2.48M paired frames), it reaches 3.66 cm held-out MPJPE (CI $[3.46, 3.86]$; G1 specialist; 2.9–6.0 cm shared across all five

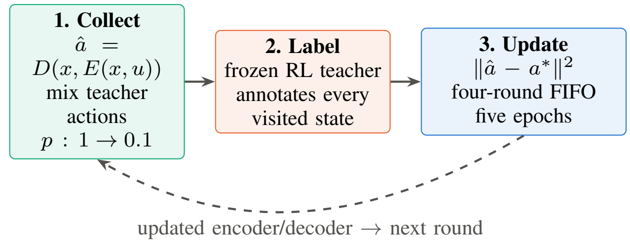

Fig. 2. **One online-distillation round (×2,000).** A deterministic student collects on its deployment path while frozen-teacher actions are mixed in with $p_r = \max(0.1, 1 - r/200)$. The teacher labels every visited state; encoder and decoder minimize action MSE using the current and previous three rounds. At deployment there is no teacher: $z_{\mathrm{cmd}} = E(x, u)$ and $a = D(x, z_{\mathrm{cmd}})$.

robots) at 671 fps on CPU, with zero joint-limit violations by construction. Foot skate (0.29 vs teacher 0.05 m/s) remains a measured fidelity gap.

The reported instrument (Fig. 2) is deterministic. The **encoder** $E_\theta : (x, u) \mapsto z_{\mathrm{cmd}}$ sees 90-d proprioception and only the arm-specific upstream signal—the current 64-d explicit goal or the two-sample 256-d SNMR latent window. For the latter, a learned projection maps 256 to 64 dimensions before a 512–256 MLP emits z_{cmd}. The **decoder** is a 512–256–128 MLP that maps (x, z_{cmd}) to 29 PD targets at 50 Hz and never sees u. Gradients from teacher-action MSE update both modules; no RL gradient enters the student.

Rollouts mix the routed specialist teacher with the student at probability $p_r = \max(0.1, 1 - r/200)$, and the frozen teacher labels every visited state. Each of 2,000 rounds collects 24 steps in 1,024 environments, then runs five Adam epochs on a four-round FIFO buffer. A fixed 4,096-sample round-0 validation split selects the lowest teacher-action-loss checkpoint after round 50; rollout completion never selects a checkpoint. A controlled calibration ablation chose this simple model: a deterministic 64-d explicit-goal encoder reached 0.969 completion, within the earlier CVAE variant's range. Posterior, KL, sampling noise, and smoothness are therefore not ingredients of E70. Earlier posterior-path and goal-free variants collapsed at deployment; they motivated deployment-path collection and goal conditioning but are not pooled with the reported assay.

IV. Experiments: From a Clock Confound to a Paired Assay

We evaluate deterministic deployment on a simulated Unitree G1 in MuJoCo-Warp [10]. *Completion* is the fraction of 10-s rollouts that remain inside the standard tracking tube: anchor drift ≤ 0.5 m, body-orientation error ≤ 0.8 rad, and end-effector deviation ≤ 0.25 m. Every reported student evaluation uses 1,024 rollouts and seed 404. Students use the same 2,000-round online-distillation recipe: 1,024 environments, 24-step collection, four-round replay, five epochs, Adam 3×10^{-4}, and validation-selected checkpoints. All results postdate the two measurement repairs in §VII.

A. Diagnostic 1: a single trajectory is a clock

On LAFAN1 [9] `walk1_subject5`, an explicit-command student matches its evaluated teacher checkpoint (completion 0.955 ± 0.008 vs 0.954; three student seeds, one teacher checkpoint). A frozen human-side z_{ret} achieves 0.656 ± 0.044 with no robot-space reference, far above a proprio-only encoder at 0.001 ± 0.001. Yet a fixed random projection of absolute frame-index sinusoids, passed through the same command path, reaches 0.754 ± 0.010. Because time uniquely identifies a deterministic clip, this null retires any single-clip claim that the latent supplies usable content beyond time; it does not bound information in the latent. Adding z_{ret} beside the explicit goal is also null (0.956 ± 0.003), consistent with a stronger redundant channel being used instead.

B. Diagnostic 2: matched presents with different futures

We froze a policy-independent two-trajectory assay before training any student. From the 77-clip LAFAN1 corpus, a reference-only screen paired 75 candidates with the fixed passing `walk1_subject5` anchor; the anchor and a previously closed candidate were excluded. Current perfect-tracking states are standardized joint positions and velocities; a pair is eligible when current RMS distance is at most 0.75 pooled SD but its 1-s future distance (11 samples) is at least 0.75 SD. Starts must be 0.5 s apart and retain a full same-clip 10-s rollout. The frozen selector ranked passing candidates first by lowest reference difficulty, then by window count and future divergence. It chose `walk1_subject1`: 69 pairs, median future distance 1.113 SD. Its specialist and the anchor specialist independently pass the same 1,024-rollout gate (0.987 and 0.847 completion; macro 0.917). The search is policy-independent but intentionally favors a feasible tracking pair.

The five student arms share the teacher ensemble, student core, training budget, checkpoint rule, and evaluation starts; they differ in the upstream signal available to the encoder: explicit robot goal (C), frozen SNMR latent at t and $t{+}0.1$ s (A), absolute within-clip time (T), proprioception alone (B), or the other clip's latent at matched normalized time (S). C runs first and must reach 0.80 general completion or fall no more than 0.05 below the specialist macro. Only then are A/T/B/S licensed. The primary endpoint is ambiguity-start completion; a paired bootstrap clusters by the 69 frame pairs, and the registered content gate requires A–T ≥ 0.10, A–T and A–S intervals excluding zero, and a positive A–T direction on both clips. T uses 16 log-spaced sine/cosine pairs followed by one fixed seed-63 random projection to 128 dimensions; the code resets identically at each clip boundary, contains no motion ID or global offset, and uses the same two-frame window as A.

Each of the 69 pair identities occurs on both trajectory sides and is repeated across the 1,024 rollout batch. We first average the paired binary completion difference within pair identity, then bootstrap the 69 pair means 10,000 times; this prevents frequently repeated starts from dominating uncertainty. The registered seed-0 analysis is immutable. For the two confirmation seeds, the final analyzer resamples training seed and pair identity in two stages, reports each seed separately, and fail-closes if any arm differs in start steps, pair IDs, side assignment, or evaluation seed. Checkpoints are chosen only by the fixed held-out teacher-action loss, never by rollout completion.

The training-seed contrasts are A–T $+0.154$, $+0.278$, and $+0.140$, and A–S $+0.187$, $+0.131$, and $+0.281$ for seeds 0–2, respectively. These values expose seed consistency rather than relying on the aggregate alone.

Table I reports PASS for the registered explicit capability gate and PASS for the registered content gate. The per-clip A–T estimates are $+0.191$ [0.129, 0.265] and $+0.190$ [0.094, 0.296]. The explicit student verifies task and trainer feasibility, and destruction of its exclusive z_{cmd} verifies causal command use. The registered contrasts show that neither absolute progress through a clip nor the matched-phase other-trajectory control is sufficient to match A. Under this controller and two-walk assay, the frozen SNMR code therefore provides control-usable trajectory information beyond absolute within-clip time. The tested comparison is substitutional: the code

TABLE I
FROZEN TWO-WALK THREE-SEED AGGREGATE (1,024 ROLLOUTS PER ARM/SEED). AMB.: STARTS FROM 69 REFERENCE-ONLY STATE-MATCHED PAIRS.

Encoder input	General	Amb.	Amb. survival
Explicit goal (C)	0.923	0.973	9.890 s
Frozen z_{ret} (A)	0.699	0.754	8.726 s
Absolute time (T)	0.579	0.562	7.353 s
Matched-phase shuffle (S)	0.536	0.552	7.448 s
Proprioception only (B)	0.431	0.437	6.833 s
A–T, paired 95% CI		$+0.191$	$[0.124, 0.274]$
A–S, paired 95% CI		$+0.199$	$[0.127, 0.279]$

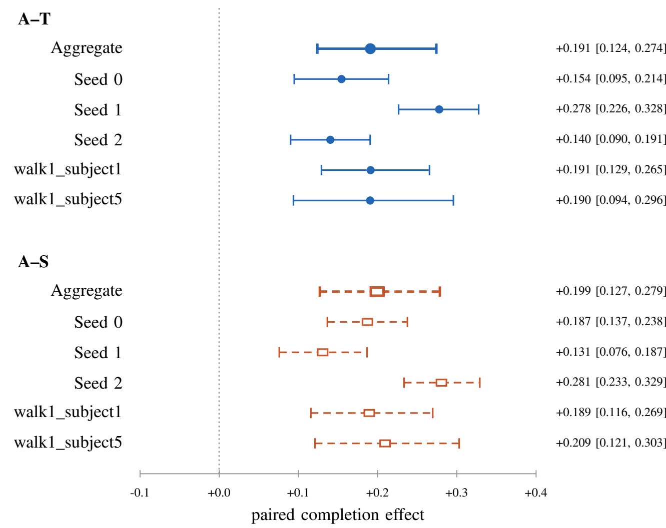

Fig. 3. **Paired E70 effects.** Circles/solid intervals are A–T; squares/dashed intervals are A–S. Aggregate rows reproduce the frozen analyzer's hierarchical intervals. Seed and clip rows are generated from the analyzer's hash-bound rollout inputs with the same 69-pair clustered bootstrap; they are descriptive and do not change the registered gate.

replaces explicit targets; its additive value beside an explicit goal remains null on the single-clip diagnostic.

V. What the Upstream Latent Exposes

Before asking the latent to carry control, we measured what it holds. Four pre-specified probe families, in increasing severity:

Content: instance-aligned, not linearly semantic. Cross-embodiment *window* retrieval hits 75–83% top-1 at 1.02% chance on four embodiments and 57% on the fifth (the morphological outlier); CKA between per-robot encodings is 0.86–0.92. A *linear* motion-category probe is near chance (0.151 vs 0.125). We therefore claim only that motion category is not linearly accessible; exploratory low-dimensional visualizations do not establish nonlinear decodability.

Embodiment: aligned, not invariant. A linear probe recovers the source embodiment at only 0.28 accuracy, but a nonlinear attacker reaches 0.91—the classic gap between "not linearly decodable" and "absent." A height-normalization intervention moves the attacker only $\sim$1 pp, suggesting—though, being an out-of-distribution intervention, not proving—that body scale is a minor share of the leak.

Physics: weakly exposed. Foot-contact state is poorly linearly decodable under pure distillation: F1 is 0.088 at full budget, below the deployable source-height heuristic at 0.127, and AUROC is 0.51–0.64 on the only clip with balanced label support. A co-trained contact head raises latent-probe F1 to 0.227 at full budget (1.8$\times$ the heuristic) but costs $+1.5$ cm MPJPE at loss weight 0.5. Contact thus becomes substantially more accessible only when it is explicitly supervised, with a measurable fidelity tradeoff.

Control: useful when exclusive, redundant beside explicit targets. Through the distilled encoder of §IV, adding the latent beside the explicit reference is null (0.956 vs 0.955). This is consistent with the explicit channel dominating when it remains available. When the latent is exclusive and two trajectories make time ambiguous, its instance-aligned structure becomes control-usable (§IV). The probes show that the frozen latent exposes trajectory identity and pose evolution, but little linearly accessible semantics or contact physics; the controller assay separately tests whether the decoder can use that content.

VI. Upstream Retargeter Boundary

The assay uses a nontrivial upstream representation rather than an arbitrary motion embedding. SNMR amortizes an IK teacher across five humanoids at 671 fps on CPU, with 3.66 cm G1 held-out MPJPE and joint limits enforced by construction. Adding 1,938 interaction-rich OmniRetarget clips [2] tests whether the shared model can absorb a second teacher and new source skeletons. Table II evaluates free-standing root prediction on group-held-out sibling splits: the shared model remains within 2 cm of each specialist in-domain. Each specialist degrades sharply off-domain (25.9–58.6 cm versus 4.1–11.5 cm in-domain). With one training seed per arm, a balanced sampling diet accounts descriptively for 61% of the interference reduction over a naive shared run.

TABLE II

Two-teacher absorption: held-out *free-root* world-frame MPJPE (cm; root predicted, not given), group-held-out sibling splits, matched budget, one training seed per arm.

Held-out pool	LAFAN1-only	Omni-only	Two-teacher
Locomotion	**4.1**	25.9	5.9
Object inter.	58.6	**11.5**	12.2
Terrain inter.	52.6	**10.8**	11.9

The representation's explicit-reference quality is also trackable. Matched policies trained on SNMR versus IK references reach 86.7% versus 86.7% completion (three training seeds, two evaluation seeds) with SNMR-minus-GMR joint-RMSE difference $+0.33\%$, CI $[-3.1, 4.9]\%$. The completion difference is 0.0 pp, CI $[-5.8, +6.2]$ pp, so formal noninferiority at a -5 pp margin is unresolved. The assay is nevertheless sensitive to reference quality: adding i.i.d. 0.05-rad joint noise drops completion from 0.94 to 0.48, nine times the prespecified detection gate. These upstream results characterize scale and fidelity; the paired interface assay, not retargeting MPJPE, establishes downstream control-usable content.

VII. A Defect in the Measurement Substrate

Calibration found two measurement defects before the reported assay was frozen. First, MuJoCo-Warp state tensors include the world body while the corresponding name list does not; an off-by-one in our pinned stack made the body-position reward exactly zero. Correcting the index cut joint RMSE from 0.263 to 0.142 rad without changing the learning algorithm. Second, observation terms are concatenated alphabetically, while our first distillation implementation sliced them in configuration order. The decoder therefore received an explicit-goal shortcut inside its nominal proprioception; blanking the leaked dimensions collapsed completion from 0.93 to 0.00. We replaced positional slices with name-derived indices and regression-test the decoder's exact 90-d proprioceptive contract. Every interface result in this paper was retrained after both repairs. The validity lesson is methodological: structural channel assertions require layout tests, and structural exclusivity still requires a causal intervention.

VIII. Reproducibility Contract

The reproducibility record separates small immutable evidence from generated training outputs. It includes the SNMR package, interface trainer, name-derived observation layout tests, reference-only selector, launchers, paired analyzer, exact environment pins, and machine-readable protocol records. The E70 record freezes ordered motion hashes, both specialist checkpoint and evaluation-report hashes, the transformed 69-pair start grid, student recipe, seeds, gates, and outcome language. The launcher uses a new E70 root, refuses either failed specialist, evaluates C before any representation arm, and writes reports atomically; completed cells are hash-checked and skipped on resume. Each report stores all 1,024 start steps, completion indicators, survival and error vectors, motion

IDs, teacher checkpoints, arm configuration, and evaluation seed. The analyzer verifies exact pairing before computing a result and records SHA-256 for every input in its output. Large LAFAN1/GMR-derived motions, teacher checkpoints, and student checkpoints are excluded from the review PDF but have deterministic preparation commands; no external artifact is required to understand the claims in Table I.

IX. Limitations

The positive content result is deliberately narrow: two similar LAFAN1 walks, one simulated Unitree G1, one frozen SNMR endpoint, one specialist ensemble, and one distilled controller class. Both evaluation motions are student-training trajectories: the ambiguity starts change the start distribution but do not test held-out-motion generalization. The matched time null rules out absolute progress through these clips, not every temporal or phase model; the shuffled control destroys clip identity as well as future correctness, so a positive result establishes clip-disambiguating information but does not separate static clip identity from finer time-varying future semantics. Nearby starts can also have overlapping 10-s rollout horizons; the preregistered interval is conditional on these two trajectories and 69 reference-defined starts and does not model temporal dependence across pair identities. A secondary temporal-block or non-overlapping-window analysis is therefore needed to assess robustness to that dependence without changing the registered primary analysis. The primary table is the preregistered three-seed aggregate. The assay establishes control-usable information, not an information-theoretic property of z_{ret}. No outcome shows additive benefit beside an explicit target. The upstream SNMR encoder is frozen, so no tracking gradient improves what it exposes. Zero-shot decoding to an unseen robot fails ($5.2\times$ error), contact is weak without explicit supervision, object and terrain interaction are absent from the tracking loop, and no hardware or sim-to-real result is reported.

X. Conclusion

The retarget-to-track boundary becomes measurable when a learned command is the controller's only route to the goal. A single-clip test first gives the correct negative diagnosis: SNMR loses to an absolute-time control. The paired two-walk assay breaks the one-to-one link between time and target trajectory and finds a registered control advantage over absolute time and the matched-phase other-trajectory control. Command-destruction controls verify the structurally exclusive instrument's causal use when the explicit student is capable. The contribution is not a universal latent or a hardware-capability claim, but a controlled experimental contract for asking what a retargeting representation actually supplies to control—and a scoped demonstration that, on these two known walks, it supplies clip-disambiguating trajectory information beyond absolute within-clip time.

Acknowledgment

OpenAI Codex and Anthropic Claude assisted with drafting and language editing, code review, and experiment orchestration. They did not generate experimental observations; the authors designed the protocols, verified the recorded artifacts, and take responsibility for all claims.

References

[1] D. Müller et al., "ReActor: Reinforcement learning for physics-aware motion retargeting," *ACM TOG*, 2026.
[2] L. Yang et al., "OmniRetarget: Interaction-preserving data generation for humanoid whole-body loco-manipulation and scene interaction," *ICRA*, 2026.
[3] J. P. Araujo et al., "Retargeting matters: General motion retargeting for humanoid motion tracking," *ICRA*, 2026.
[4] Q. Liao et al., "BeyondMimic: From motion tracking to versatile humanoid control via guided diffusion," arXiv:2508.08241, 2025.
[5] Z. Chen et al., "GMT: General motion tracking for humanoid whole-body control," arXiv:2506.14770, 2025.
[6] K. Yin et al., "UniTracker: Learning universal whole-body motion tracker for humanoid robots," *IEEE Robot. Autom. Lett.*, 2026.
[7] X. B. Peng et al., "DeepMimic: Example-guided deep reinforcement learning of physics-based character skills," *ACM TOG*, 2018.
[8] S. Ross, G. Gordon, and D. Bagnell, "A reduction of imitation learning and structured prediction to no-regret online learning," *AISTATS*, 2011.
[9] F. G. Harvey et al., "Robust motion in-betweening," *ACM TOG*, 2020.
[10] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," *IROS*, 2012.
[11] Z. Luo et al., "Universal humanoid motion representations for physics-based control," *ICLR*, 2024.
[12] C. Tessler et al., "MaskedMimic: Unified physics-based character control through masked motion inpainting," *SIGGRAPH Asia*, 2024.
[13] H. Yao et al., "ControlVAE: Model-based learning of generative controllers for physics-based characters," *SIGGRAPH Asia*, 2022.
[14] T. He et al., "HOVER: Versatile neural whole-body controller for humanoid robots," *ICRA*, 2025.
[15] M. Ji et al., "ExBody2: Advanced expressive humanoid whole-body control," arXiv:2412.13196, 2024.
[16] X. Chen et al., "A scalable whole-body motion transfer via implicit kinodynamic motion retargeting," arXiv:2509.15443, 2025.
[17] Q. Zhao et al., "Make tracking easy: Neural motion retargeting for humanoid whole-body control," arXiv:2603.22201, 2026.
[18] H. Zhang et al., "AdaMorph: Unified motion retargeting via embodiment-aware adaptive transformers," arXiv:2601.07284, 2026.
[19] W. Zeng et al., "Behavior foundation model for humanoid robots," arXiv:2509.13780, 2025.
[20] Y. Li et al., "BFM-Zero: A promptable behavioral foundation model for humanoid control using unsupervised reinforcement learning," *ICLR*, 2026.
[21] Z. Li et al., "From language to locomotion: Retargeting-free humanoid control via motion latent guidance," *ICLR*, 2026.
[22] X. He et al., "ULTRA: Unified multimodal control for autonomous humanoid whole-body loco-manipulation," arXiv:2603.03279, 2026.
[23] Z. Luo et al., "SONIC: Supersizing motion tracking for natural humanoid whole-body control," arXiv:2511.07820, 2025.
[24] Z. Qi et al., "Humanoid-GPT: Scaling data and structure for zero-shot motion tracking," *CVPR*, 2026.
[25] S. Li, S. Li, J. Li, and M. Ding, "AnyBody: Free-form whole-body humanoid control from arbitrary keypoint guidance," arXiv:2606.29209, 2026.
[26] L. Yang et al., "HANDOFF: Humanoid agentic task-space whole-body control via distilled complementary teachers," arXiv:2606.06493, 2026.
[27] F. Amadio and E. Mingo Hoffman, "What matters in humanoid general motion tracking? An empirical study," arXiv:2607.19903, 2026.
[28] Y. Li et al., "OmniTrack: General motion tracking via physics-consistent reference," arXiv:2602.23832, 2026.
[29] Z. Ling et al., "GenTrack: Physical alignment for robot-native motion generation and zero-shot humanoid tracking," arXiv:2608.01410, 2026.
[30] Z. Meng et al., "Distinguishing imitation error from intrinsic motion learning difficulty," arXiv:2512.07248, 2025.

[31] T. Wang et al., "HumanoidArena: Benchmarking egocentric hierarchical whole-body learning," arXiv:2606.17833, 2026.
[32] G. Alain and Y. Bengio, "Understanding intermediate layers using linear classifier probes," arXiv:1610.01644, 2016.
[33] Y. Belinkov, "Probing classifiers: Promises, shortcomings, and advances," *Computational Linguistics*, vol. 48, no. 1, 2022.
[34] Y. Elazar, S. Ravfogel, A. Jacovi, and Y. Goldberg, "Amnesic probing: Behavioral explanation with amnesic counterfactuals," *Trans. Assoc. Comput. Linguistics*, vol. 9, 2021.
[35] Y. Xu, S. Zhao, J. Song, R. Stewart, and S. Ermon, "A theory of usable information under computational constraints," *ICLR*, 2020.
[36] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor, "The preregistration revolution," *PNAS*, vol. 115, no. 11, 2018.
[37] L. Bertinetto, J. F. Henriques, S. Albanie, M. Paganini, and G. Varol, eds., *Proc. NeurIPS 2020 Workshop on Pre-registration in Machine Learning*, PMLR vol. 148, 2021.
[38] R. Geirhos et al., "Shortcut learning in deep neural networks," *Nature Machine Intelligence*, vol. 2, 2020.